

Teacher's Name _____



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Student Number

Knox Grammar School 2012 Year 11 Yearly Examination

Mathematics

General Instructions

- Reading Time – 5 minutes
- Working Time – 3 hours
- Write using black or blue pen
- Board approved calculators may be used
- Answer in simplest exact form unless otherwise stated
- Show all necessary working in questions 11 – 16

Total Marks – 100

Section I 10 Marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II 90 marks

- Attempt Questions 11 – 16

Examiner: E. Choy

Number of Students in Course: 130

**This paper MUST NOT be removed from
the examination room**

Section I Multiple Choice

10 Marks

Attempt Question 1 – 10.

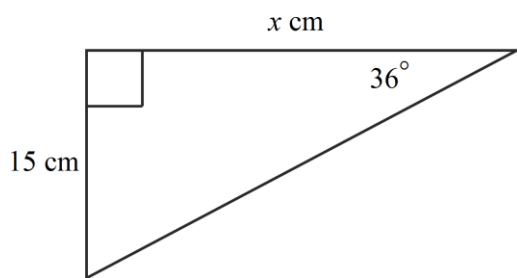
Allow approximately 15 minutes for this section.

1. Evaluate $\frac{3.23}{0.45+1.2^2}$ correct to 3 significant figures.
- (A) 1.71 (B) 8.62
(C) 8.617 (D) 1.708
2. If $\frac{2}{2-\sqrt{3}} = a + \sqrt{b}$ then the values of a and b are:
- (A) -4 and 12 (B) 4 and 12
(C) 4 and 3 (D) 2 and 3
3. The solutions to the equation $2x^2 - 7x - 2 = 0$ are:
- (A) $\frac{-7 \pm \sqrt{33}}{4}$ (B) $\frac{-7 \pm \sqrt{65}}{4}$
(C) $\frac{7 \pm \sqrt{33}}{4}$ (D) $\frac{7 \pm \sqrt{65}}{4}$
4. The domain and range for the function $y = \sqrt{7-x}$ is:
- (A) $x \leq 7$; $y \geq 0$ (B) $0 \leq x \leq 7$; $y \geq 0$
(C) $x \geq 0$; $y \geq 0$ (D) All real x , All real y
5. A function is defined by $f(x) = x^2 - 4x + 5$.
The point on $y = f(x)$ which has a gradient of 2 is:
- (A) $(3, -1)$ (B) $(2, 1)$
(C) $(3, 2)$ (D) $(2, 3)$

6. The derivative of $2x^{\frac{1}{2}}$ is :

- (A) $-\frac{2}{\sqrt{x}}$ (B) $-\frac{1}{\sqrt{x}}$
 (C) $\frac{1}{\sqrt{x^3}}$ (D) $\frac{1}{\sqrt{x}}$

7. The calculation which would be used to find the value of x is:



- (A) $15 \tan 36^\circ$ (B) $15 \cos 36^\circ$
 (C) $\frac{15}{\tan 36^\circ}$ (D) $\frac{15}{\cos 36^\circ}$

8. The equation $3x^2 + 2x - 1 = 0$ has roots α and β . The value of $2\alpha + 2\beta$ is :

- (A) $-\frac{4}{3}$ (B) $-\frac{2}{3}$
 (C) $-\frac{1}{3}$ (D) 10

9. The equation $x^2 + y^2 - 6x + 6 = 0$ represents a circle. The coordinates of the centre of the circle are:

- (A) (3, -3)
 (B) (-3, -3)
 (C) (3, 0)
 (D) (-6, 0)

10. For what values of k is the expression $-\frac{x^2}{4} - x - k$ negative definite ?

- (A) $k < 1$ (B) $k > 1$
 (C) $k < 4$ (D) $k > 4$

End of Section I

Section II

90 marks

Attempt Questions 11 – 16

Allow about 2 hours 45 minutes for this section

Answer each question in a separate writing booklet. Extra writing booklets are available.

All necessary working should be shown in every question.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) Evaluate $\sqrt[3]{\frac{651}{4\pi}}$ correct to two decimal places. 2

(b) Simplify $\frac{a^2 - 25}{a - 5}$ 1

(c) (i) Draw a neat sketch of the parabola $y = (2x - 1)(x + 3)$ noting x and y intercepts. 2

(ii) Hence, determine the values of x for which $(2x - 1)(x + 3) < 0$. 1

(d) If $f(x) = x^2 - 3x + 2$ and $g(x) = x^2 - x - 4$, for what value of x is $f(x) = g(x)$? 1

(e) On separate number planes, sketch neatly each of the following curves:

(i) $y = \frac{1}{x}$ 1

(ii) $y = |x| + 1$ 1

(iii) $y = \sqrt{25 - x^2}$ 1

(f) Find the exact value of θ such that $2\cos\theta = 1$ where $0^\circ \leq \theta \leq 90^\circ$. 1

(g) Solve the pair of simultaneous equations:

$$\begin{aligned} 2x + y &= 7 \\ x - 2y &= 1. \end{aligned}$$
2

(h) What are the domain and range of $y = x^2\sqrt{1+x}$? 2

Question 12 (15 marks) Use a SEPARATE writing booklet

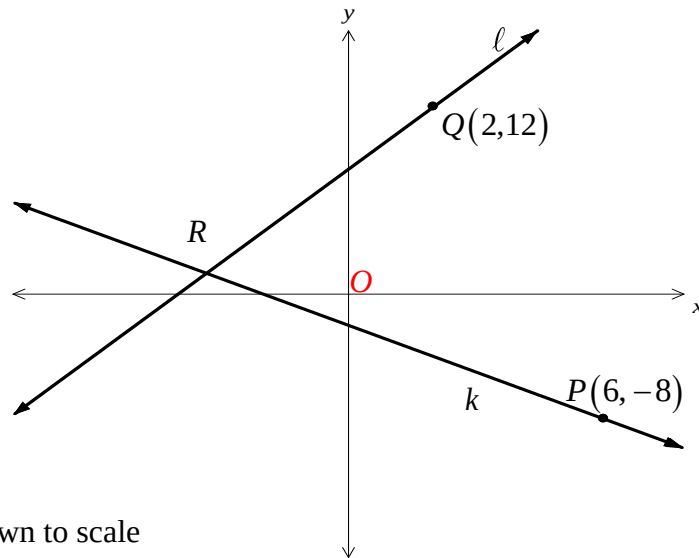


Diagram not drawn to scale

- (a) A line ℓ whose equation is $2x - y + 8 = 0$ has the point $Q(2, 12)$ on it.
 Another line k has gradient -2 and passes through the point $P(6, -8)$. The lines k and ℓ intersect at the point R .
- (i) Show that the equation of the line k is $2x + y - 4 = 0$. 1
- (ii) Show that the co-ordinates of R are $(-1, 6)$. 2
- (iii) Show that the distance of QR is $3\sqrt{5}$. 1
- (iv) Find the perpendicular distance from P to the line ℓ .
 Leave your answer in surd form. 2
- (v) Find the area of $\triangle PQR$. 1

Question 12 continues on page 6

Question 12 continued

- (b) Shade the region in the plane for which the inequations $y \leq 4 - x^2$ and $y \geq 0$ hold simultaneously. 2
- (c) Find the locus of the point $P(x, y)$ whose distance from the point $A(4, 0)$ is always twice its distance from the point $B(1, 0)$. 2
- (d) The general form of a line ℓ passing through the point of intersection of $x - 2y + 5 = 0$ and $x + y + 2 = 0$ is $(x - 2y + 5) + k(x + y + 2) = 0$.
- (i) Show that the gradient of line ℓ is $\frac{1+k}{2-k}$. 2
- (ii) Hence find the equation of the line passing through the point of intersection of $x - 2y + 5 = 0$ and $x + y + 2 = 0$ and has gradient 2. 2
Leave your answer in general form.

End of Question 12

Question 13 (15 marks) Use a SEPARATE writing booklet

- (a) Two ships set sail from a point O . The first ship sails for 45 nautical miles on a course bearing $N32^\circ W$ to a point Y and the second ship sails for 63 nautical miles on a course bearing $N58^\circ E$ to a point Z .
- (i) Draw a half-page diagram representing the above information. **1**
- (ii) Find, to the nearest nautical mile, the distance YZ . **2**
- (b) The adjacent sides of a parallelogram have lengths 4 cm and 6 cm and they include an angle of 60° . Calculate the length of the shorter diagonal to 2 decimal places. **2**
- (c) Given that $\tan \theta = -\frac{5}{12}$ and θ is obtuse, find $\sin \theta$ and $\sec \theta$. **2**
- (d) If $\sin \theta = x$, express $\frac{1 - \cos^2 \theta}{\sec^2 \theta}$ in terms of x . **2**
- (e) (i) Show that $\cos \theta \tan \theta = \sin \theta$ **1**
- (ii) Hence solve $8 \sin \theta \cos \theta \tan \theta = \operatorname{cosec} \theta$ for $0^\circ \leq \theta \leq 360^\circ$. **3**
- (f) The distance, d km, between two ports P_1 and P_2 can be found from the calculation:
- $$d^2 = 900 \left[\left(\frac{\sin 44^\circ}{\sin 26^\circ} \right)^2 + \left(\frac{\sin 19^\circ}{\sin 21^\circ} \right)^2 \right].$$
- (i) Determine the distance, d , from port P_1 to port P_2 , correct to two decimal places. **1**
- (ii) Determine the time it will take to sail from P_1 to P_2 at an average speed of 10km/h. **1**
Give your answer to the nearest minute.

Question 14 (15 marks) Use a SEPARATE writing booklet

(a) If $g(x) = x^2 + 3x + 5$, show that $g(a+1) = g(a) + 2(a+2)$ 2

(b) Find $\lim_{x \rightarrow \infty} \frac{x^2 - x - 12}{x^2 - 4x}$ 1

(c) Differentiate:

(i) $x^4 + x^{-1}$ 1

(ii) $(x^2 - 1)^5$ 2

(iii) $\frac{x-3}{5x+7}$ 2

(d) Find the equation of the tangent to the curve $y = 2x^2 - 4x + 1$ at the point where $x = 1$. 2

(e) A function is defined by $f(x) = \begin{cases} x, & \text{for } -1 \leq x \leq 1 \\ x(2-x), & \text{for } 1 < x \leq 2 \end{cases}$

(i) Find $f(0) + f(1.5)$ 1

(ii) Sketch the graph $y = f(x)$ for $-1 \leq x \leq 2$. 2

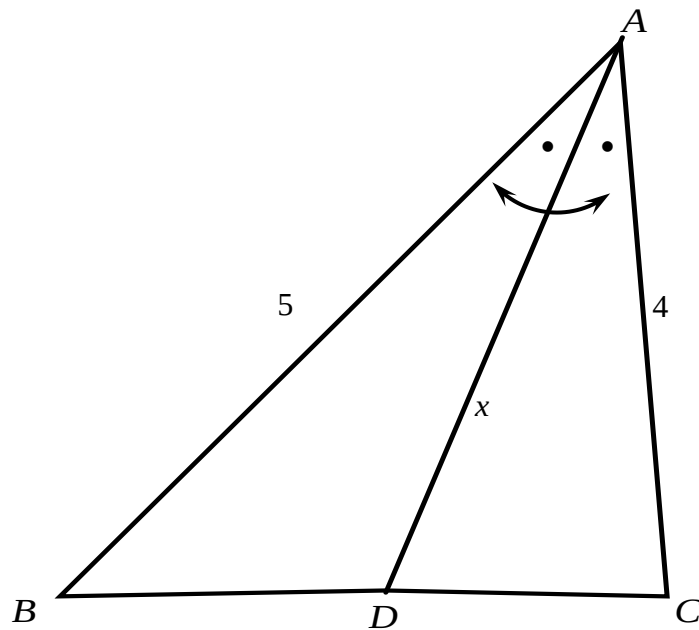
(iii) The horizontal line $y = k$ meets the graph of $y = f(x)$ *exactly once*. 2
Find the range of values of k for this to be true.

Question 15 (15 marks) Use a SEPARATE writing booklet

- (a) If α and β are the roots of the equation $2x^2 + 7x + 4 = 0$, find the value of :
- | | | |
|-------|----------------------|---|
| (i) | $\alpha + \beta$ | 1 |
| (ii) | $\alpha\beta$ | 1 |
| (iii) | $\alpha^2 + \beta^2$ | 2 |
- (b) For the parabola $(y-1)^2 = 4(x+4)$, find:
- | | | | |
|-----|-------|--|---|
| (α) | (i) | The co-ordinates of the vertex. | 1 |
| | (ii) | The co-ordinates of the focus. | 1 |
| | (iii) | The equation of the directrix. | 1 |
| | (iv) | The length of the latus rectum. | 1 |
| | (v) | Its x and y intercepts. | 2 |
| (β) | | Hence, draw a neat sketch of the parabola. | 1 |
- (c) Find the values of m for which the quadratic equation $x^2 - 3mx + 9 = 0$ has real and distinct roots. 2
- (d) If one of the roots of $x^2 + ax + b = 0$ is three times the other, prove that $16b = 3a^2$. 2

Question 16 (15 marks) Use a SEPARATE writing booklet

(a)



The above figure shows a triangle ABC such that $AB = 5\text{cm}$, $AC = 4\text{cm}$ and $\angle BAC = 60^\circ$. D is a point on BC such that AD bisects $\angle BAC$.

(i) Express the area of $\triangle ABD$ in terms of x . **1**

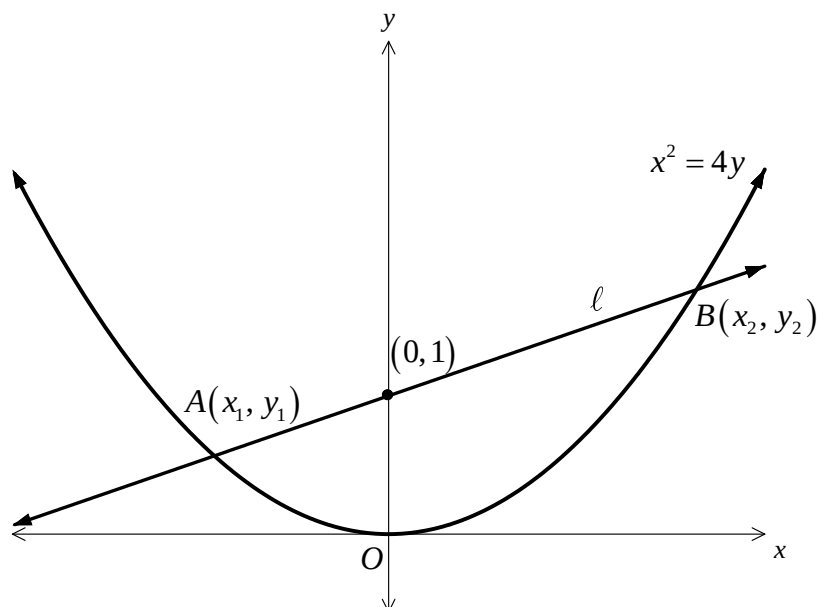
(ii) Hence, using triangles ADC and ABC , show that $x = \frac{20\sqrt{3}}{9}$. **2**

(b) For what values of m is $y = mx - 6$ a tangent to the parabola $y = x^2 - 2x + 3$? **3**

Question 16 continues on page 11

Question 16 continued

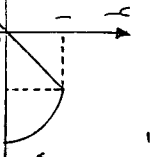
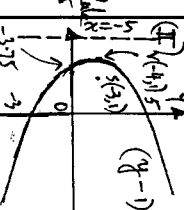
(c)



A straight line ℓ with gradient m passes through the point $(0, 1)$. The line ℓ also cuts the parabola $x^2 = 4y$ at two points $A(x_1, y_1)$ and $B(x_2, y_2)$ as shown above.

- (i) Show that the equation of the line ℓ is $y = mx + 1$. 1
- (ii) Line ℓ cuts the parabola at two distinct points. Show that x_1 and x_2 are the roots of the equation $x^2 - 4mx - 4 = 0$. 2
- (iii) Show that $(x_2 - x_1)^2 = (x_2 + x_1)^2 - 4x_1x_2$.
Hence, or otherwise find $(x_2 - x_1)^2$ in terms of m . 2
- (iv) From (iii), show that $AB = 4(1 + m^2)$. 3

End of Examination

Suggested Solution (S)	Comments	Suggested Solution (S)	Comments
<u>Question 14:</u> (a) LHS = $g(a+1) = (a+1)^2 + 3(a+1) + 5$ $= a^2 + 2a + 1 + 3a + 3 + 5$ $= a^2 + 5a + 9$ ✓ RHS = $g(a) + 2(a+2) = a^2 + 3a + 5 + 2a + 4$ $= a^2 + 5a + 9$ ✓ (b) $\lim_{x \rightarrow 0} \frac{x^2 - x - 12}{x^2 - 4x} = \lim_{x \rightarrow 0} \frac{x^2}{x^2} = 1$ ✓ LHS = RHS ✓ (c) $\frac{d}{dx}(x^4 + x^{-1}) = 4x^3 - x^{-2}$ ✓ (ii) $\frac{d}{dx}[(x^2-1)^5] = 5(x^2-1)^4 \cdot \frac{d}{dx}(x^2-1)$ $= 10x(x^2-1)^4$ ✓ (iii) $\frac{d}{dx}\left(\frac{x-3}{5x+7}\right) = \frac{(5x+7)\frac{d}{dx}(x-3) - (x-3)\frac{d}{dx}(5x+7)}{(5x+7)^2}$ $= \frac{(5x+7)(1) - (x-3)(5)}{(5x+7)^2}$ ✓ $= \frac{-22}{(5x+7)^2}$ ✓ (d) Gradient function, $\frac{dy}{dx} = 4xy - 4$ = 0 when $x=1$ No gradient is zero, line is horiz. So eqn of line is $y=-1$ (e) (i) $f(0) + f(1-5) = 0 + 15 \times 0.5 = 0.75$ or $\frac{3}{4}$ (ii)  (iii) $-1 \leq k < 0$, $k=1$ ✓		<u>Question 15:</u> (a) (i) $\alpha + \beta = -\frac{1}{2}$ (ii) $\alpha\beta = \frac{2}{3} = 2$ ✓ (iii) $\alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta$ $= (-\frac{1}{2})^2 - 2 \times 2$ $= \frac{1}{4} - 4 = -\frac{15}{4}$ ✓ (b) (i) x intercept corresponds to $y=0 \Rightarrow x = -\frac{15}{4}$ or -3.75 y intercepts correspond to $x=0 \Rightarrow y^2 - 2y - 15 = 0 \Rightarrow (y-5)(y+3) = 0$ $\Rightarrow y = -3$ or 5 (ii) $(y-1)^2 = 4(x+4)$ ✓ slope  ✓ features (c) $(-3m)^2 - 4 \times 1 \times 9 > 0 \Rightarrow 9m^2 - 36 > 0$ $\Rightarrow m^2 - 4 > 0 \Rightarrow (m+2)(m-2) > 0$ ✓ So $m < -2$ or $m > 2$ ✓ Equivalently (d) Let α, β be the roots with $\alpha > \beta$. Then $\alpha + \beta = -a$, i.e., $4\alpha = -a$ or $\alpha = -\frac{a}{4}$ Also $\alpha\beta = b$, i.e., $3\alpha^2 = b$. Sub. $\alpha = -\frac{a}{4}$ $\Rightarrow 3 \times (\frac{a^2}{16}) = b$ or $3a^2 = 16b$, as required ✓ <u>Question 16:</u> (a) (i) Area of $\triangle ABC$ given by $A = \frac{1}{2} \times 5 \times x \sin 30^\circ = \frac{5x}{4}$ ✓ (ii) Similarly, area of $\triangle ADC$ is: $A = \frac{1}{2} \times 4 \times x \sin 30^\circ = x \text{ cm}^2$ & $A_1(\triangle ABC) + A_2(\triangle ADC) = A_1(\triangle ABC)$, i.e., $\frac{5x}{4} + x = \frac{1}{2} \times 5 \times 4 \times \sin 60^\circ \Rightarrow \frac{9x}{4} = 5\sqrt{3}$ or $x = \frac{20\sqrt{3}}{9}$, as required. (b) Solving simultaneously $\begin{cases} y = mx - 6 \\ x^2 - 2x + 3 = mx - 6 \end{cases} \Rightarrow x^2 - 2x + 3 = mx - 6$ $\Rightarrow x^2 - (m+2)x + 9 = 0$ Lagrange $\Delta = 0$ $\Rightarrow [-(m+2)]^2 - 4 \times 1 \times 9 = 0$ No solution if points given in lieu of intercepts: 2 marks for all three intercepts. $(y-k)^2 = 4a(x-h)$ $\Delta > 0$ i.e., $b^2 - 4ac > 0$	

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
<u>Question 16 cont.</u> (i) From $y = mx + b$ it is clear the y-intercept is 1 from the diagram. Also $y = mx + 1$ is required (ii) Solving $\begin{cases} x^2 = 4y \\ y = mx + 1 \end{cases}$ simultaneously $\Rightarrow x^2 = 4(mx + 1)$ $\Rightarrow x^2 - 4mx - 4 = 0$. However, from graph we know that the x coordinate of the points of intersection of the graphs are x_1 & x_2. Hence x_1, x_2 are the roots of $x^2 - 4mx - 4$, as required (iii) $\Delta HS = (x_2 - x_1)^2$ $= x_2^2 - 2x_1x_2 + x_1^2$ $\Delta HS = (x_2 + x_1)^2 - 4x_1x_2$ $= x_2^2 + 2x_1x_2 + x_1^2 - 4x_1x_2$ $= x_2^2 - 2x_1x_2 + x_1^2$ $= \Delta HS$ From (ii) $x_1 + x_2 = 4m$ & $x_1x_2 = -4$ $\therefore (x_2 - x_1)^2 = (4m)^2 - 4(-4)$ $= 16m^2 + 16 \checkmark$	Correct Δ intercept from substituting $(0, 1)$ into the line. It is correct. It is a straight line. The form $y = mx + 1$.	(iv) From the distance formula $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ $\& x_1^2 = 4y_1$, with $x_2^2 = 4y_2$ $\Rightarrow y_2 - y_1 = \frac{1}{4}x_2^2 - \frac{1}{4}x_1^2$ $= \frac{1}{4}(x_2^2 - x_1^2) \checkmark$ $\Rightarrow AB = \sqrt{(x_2 - x_1)^2 + \left\{\frac{1}{4}(x_2^2 - x_1^2)\right\}^2}$ $= \sqrt{(x_2 - x_1)^2 + \frac{1}{16}(x_2 + x_1)^2(x_2 - x_1)^2}$ <u>Difference of Squares.</u> $= \sqrt{16m^2 + 16 + \frac{1}{16}(4m)^2(16m^2 + 16)}$ $= \sqrt{(16m^2 + 16) + m^2(16m^2 + 16)}$ $= \sqrt{(16m^2 + 16)(1 + m^2)}$ $= \sqrt{16(m^2 + 1)^2}$ $= 4(m^2 + 1)$, as required (v) If $y_1 = y_2 \Rightarrow m = 0 \checkmark$ $\therefore AB = 4$ units <u>to 2 d.p.</u>	(x_1, y_1) & (x_2, y_2) lie on parabolas $x^2 = 4y$ 2 marks for process $1/2$ for meaning full process $m^2 + 1 > 0$